MANAGEMENT SYSTEMS

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Module: principle of programing

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Date: 01/2025



AGENDA

Introduction

Hotel management system

Student management system

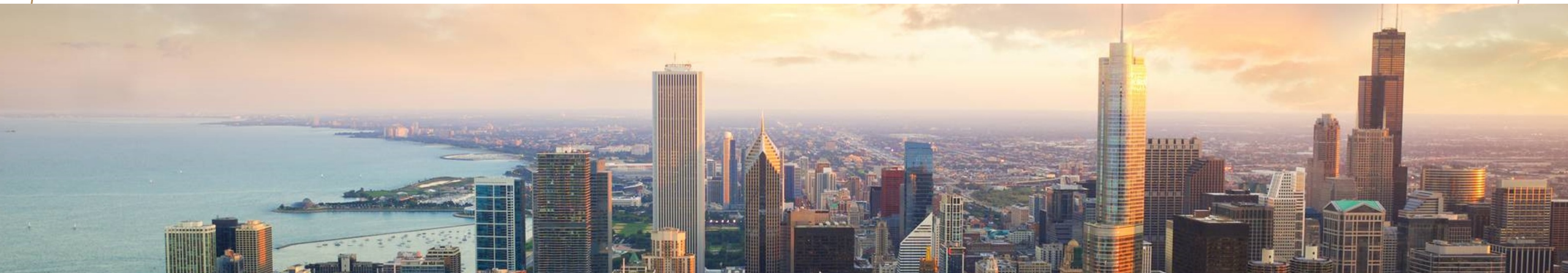
Record

Conclusion



INTRODUCTION

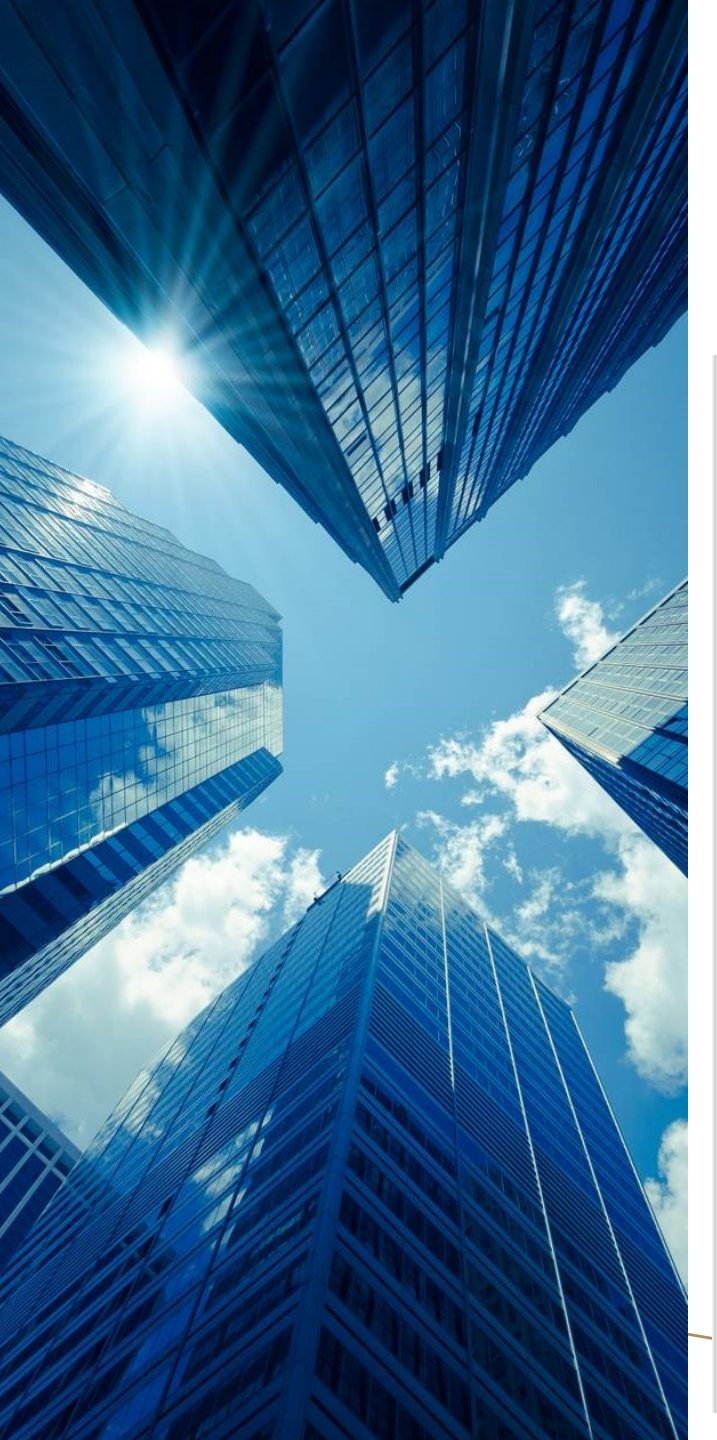
A Hotel Management System is one that carries out the automation of hotel operations in keeping the customer's information safe and easy to check and track. A Student Management System is the one that carries out automation of students' data in schools or higher education institutions, including adding, viewing, search or delete a student.



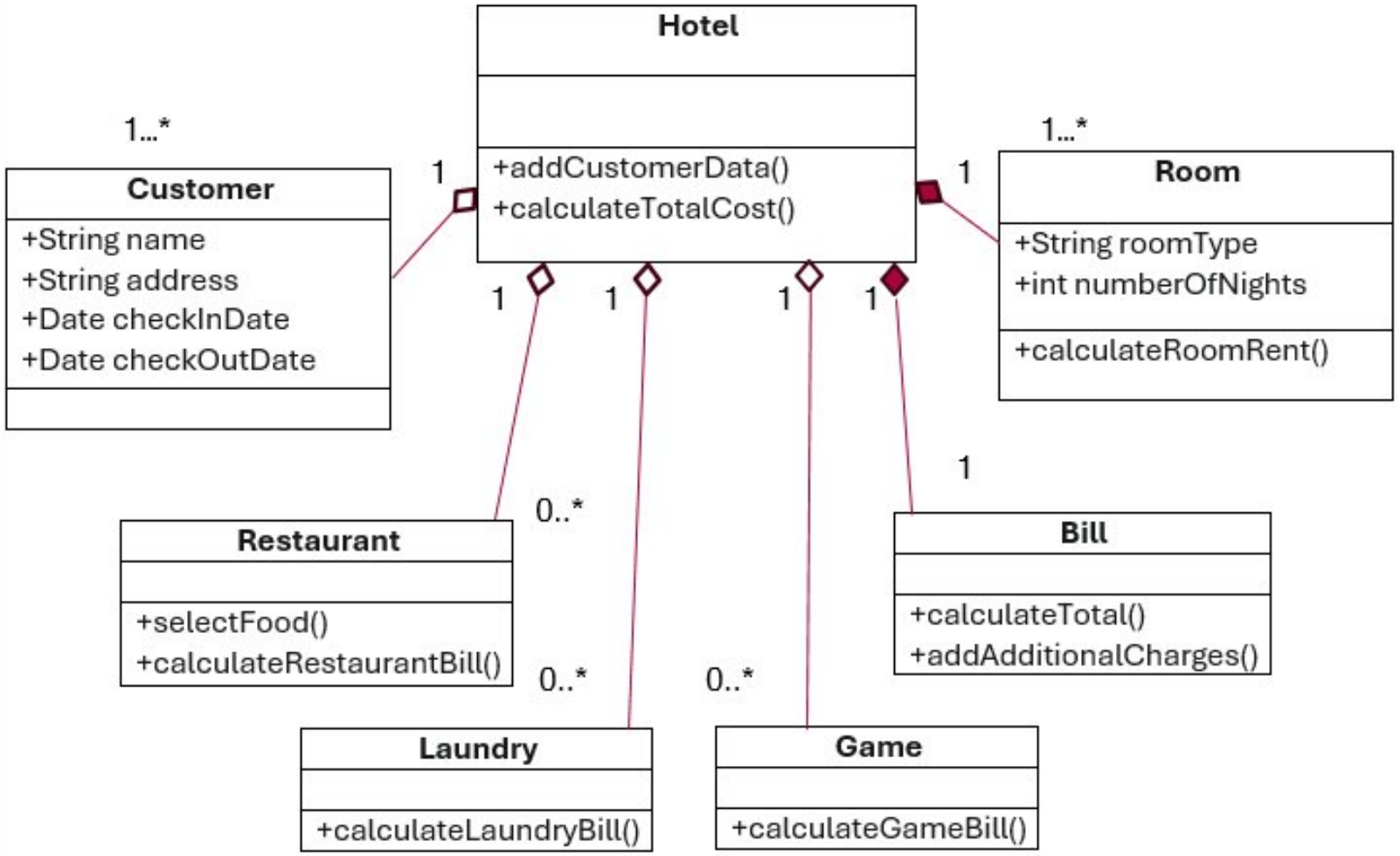
HOTEL MANAGEMENT SYSTEM

HOTEL MANAGEMENT SYSTEM IS A SIMPLE JAVA-BASED CONSOLE APPLICATION, NAMED HOTELAURORA.JAVA IT IS ORIENTED FOR EASE OF USE. IT'S USED FOR FEEDING CUSTOMER DATA, CALCULATION OF THE RENT OF THE ROOM, AND DEVELOPMENT OF BILLS FOR SERVICES AVAILED LIKE A RESTAURANT, LAUNDRY, AND GAMES. THERE IS NO LOGIN IN IT, AND NO DATABASE HAS BEEN USED, MAKING IT LIGHT AND SOLELY FOR QUICK CALCULATIONS AND EASE OF USE.

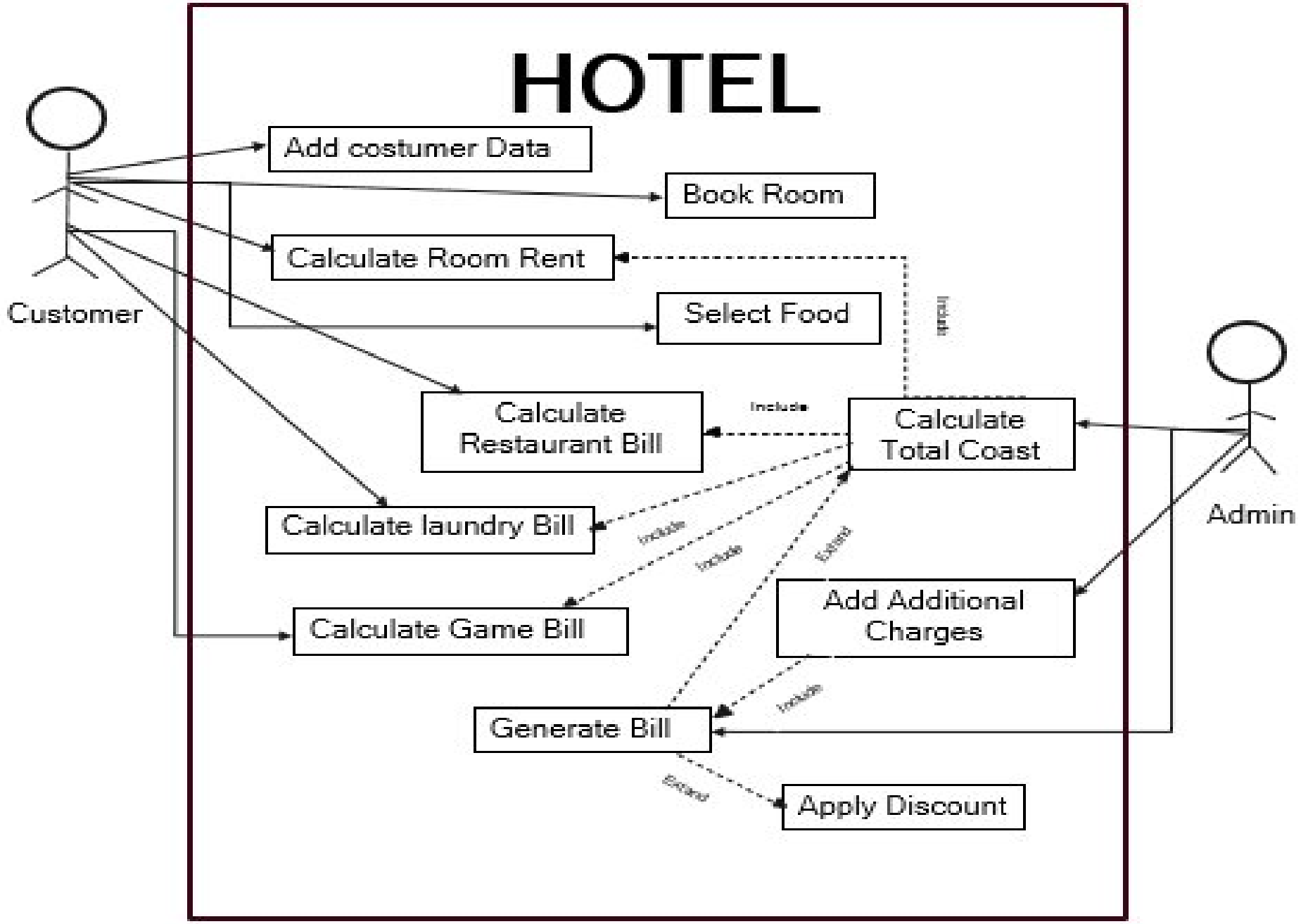




UML DIAGRAM


Hotel
+addCustomerData()
+calculateTotalCost()
Customer
+String name
+String address
+Date checkInDate
+Date checkOutDate
Room
+String roomType
+int numberOfNights
+calculateRoomRent()
Restaurant
+selectFood()
+calculateRestaurantBill()
Bill
+calculateTotal()
+addAdditionalCharges()
Laundry
+calculateLaundryBill()
Game
+calculateGameBill()
1...*
1
0..*

USE CASE


HOTEL
Customer
Add costumer Data
Book Room
Calculate Room Rent
Select Food
Calculate Restaurant Bill
Include
Calculate Total Coast
Admin
Calculate laundry Bill
Include
Include
Extend
Calculate Game Bill
Add Additional Charges
Include
Generate Bill
Extend
Apply Discount

File Edit Selection View Go Run ··· .redhat

EXPLORER

.REDHAT

anonymousId

HotelAurora.java

Studentsmanagementsystem.java

HotelAurora.java Studentsmanagementsystem.java

HotelAurora.java > HotelAurora > main(String[])

```java
public class HotelAurora {
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);
        System.out.println(x:"Welcome to Hotel Aurora");
        System.out.println(x:"Enter Full Name :");
        String name = scan.nextLine();
        System.out.println(x:"Enter Adress :");
        String adress = scan.nextLine();
        System.out.println(x:"Enter Check-in Date :");
        String checkindate = scan.nextLine();
        System.out.println(x:"Enter Check-out Date :");
        String checkoutdate = scan.nextLine();

        // Rooms type selection
        System.out.println(x:" Rooms type ");
        System.out.println(x:"1. Single Room - 100$ per night");
        System.out.println(x:"2. Double Room - 150$ per night");
        System.out.println(x:"3. Deluxe Room - 250$ per night");
        System.out.println(x:"4. Suite Room - 300$ per night");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

PORTS

No forwarded ports. Forward a port to access your locally running services over the internet.

Forward a Port

TERMINAL

```
Games Ticket cost :$6.0
Total costs: $345.0
PS C:\Users\sugar\.redhat> ^C
PS C:\Users\sugar\.redhat>
PS C:\Users\sugar\.redhat>  c:; cd 'c:\Users\sugar\.redhat'; &
ava-21-openjdk-21.0.5.0.11-1\bin\java.exe' '-XX:+ShowCodeDetai
' 'C:\Users\sugar\AppData\Roaming\Code\User\workspaceStorage\2
a9bb\redhat.java\jdt_ws\.redhat_cc155ca0\bin' 'HotelAurora'
Welcome to Hotel Aurora
Enter Full Name :
```

OUTLINE

TIMELINE

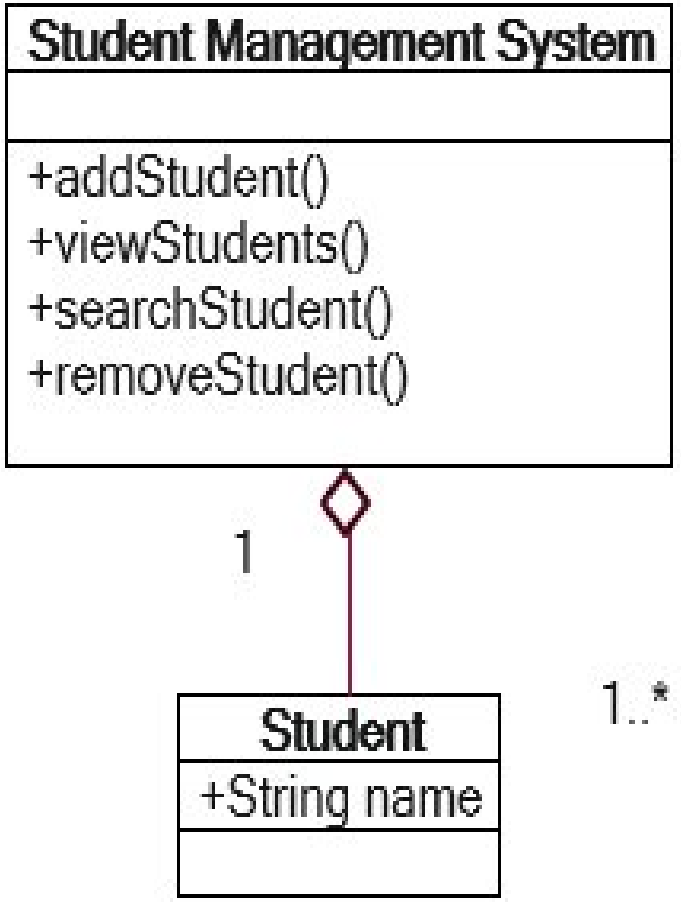
JAVA PROJECTS

0 0 0 Java: Ready

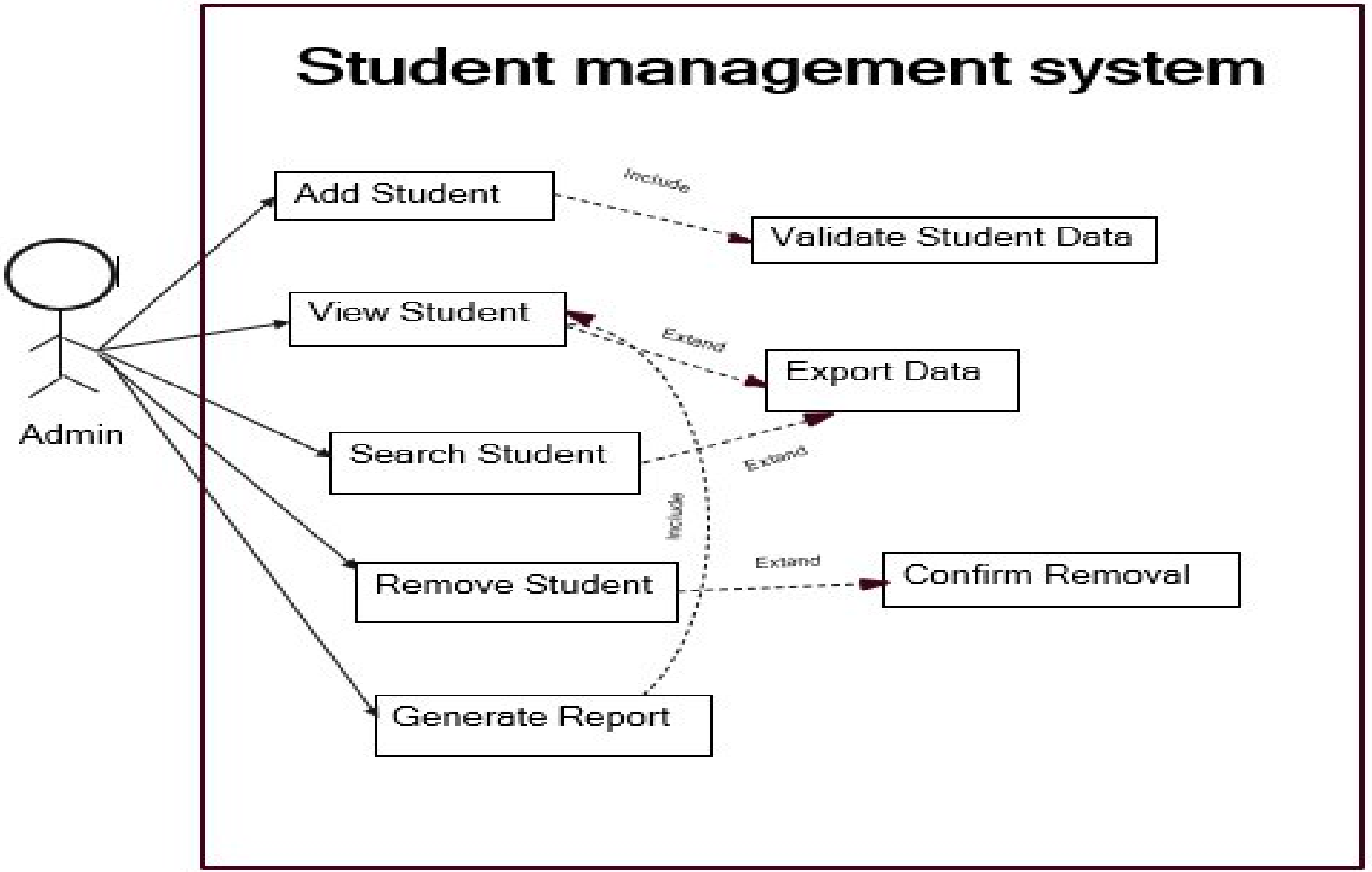
STUDENT MANAGEMENT SYSTEM

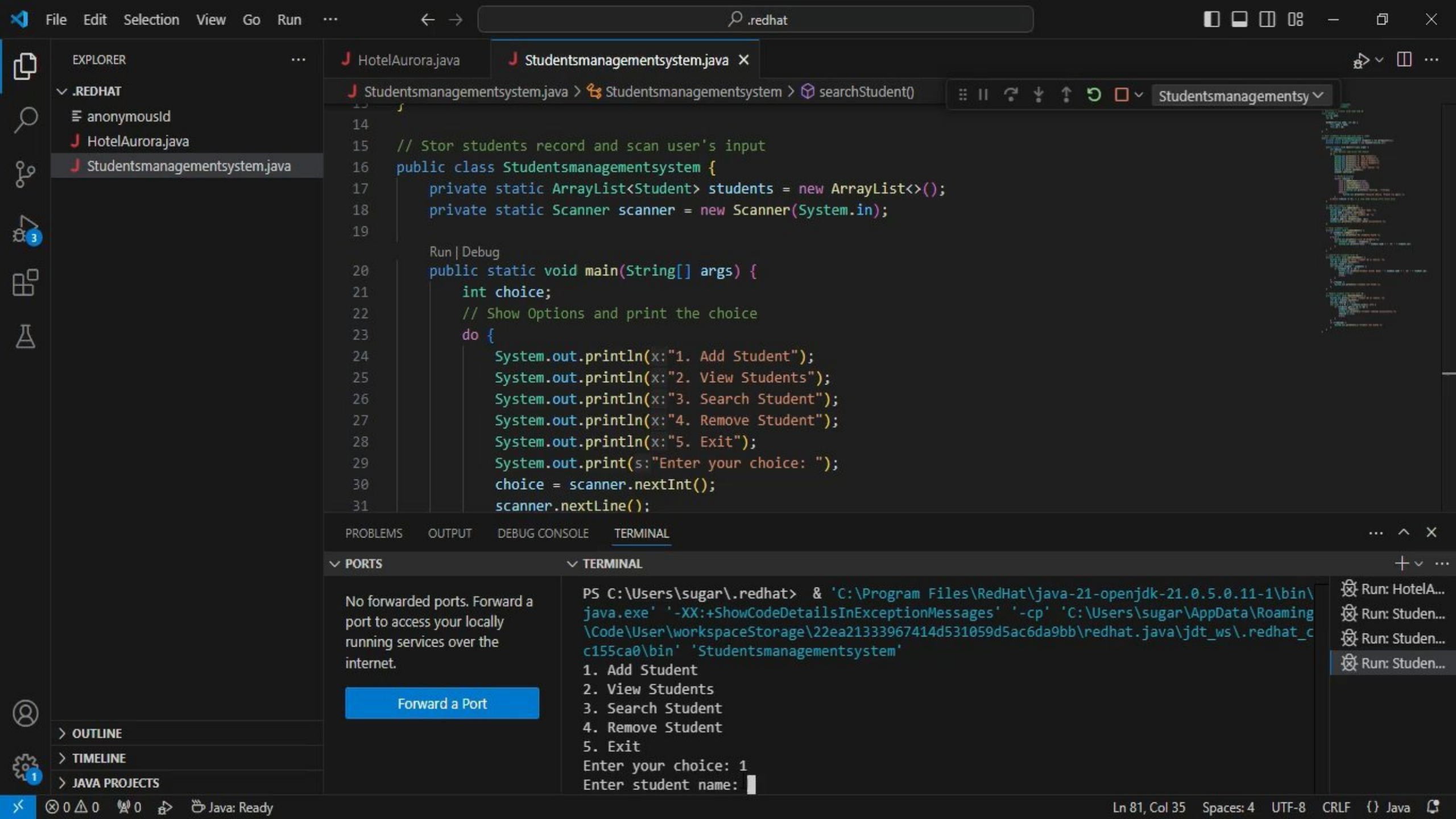
THE STUDENT MANAGEMENT SYSTEM FOLLOWS A JAVA-BASED CONSOLE APPLICATION DESIGNED TO OFFER THE USER EASY MANIPULATION: ADDING, VIEWING, SEARCHING, AND REMOVAL OF STUDENT RECORDS. THE DESIGN SCOPE IS FOCUSED ON BASIC CRUD OPERATIONS. THE APPLICATION DOES NOT REQUIRE THE USE OF A LOGIN OR DATABASE; IT IS RATHER SIMPLE AND LIGHTWEIGHT, USED FOR MAINTAINING STUDENT LISTS.

UML CLASS DIAGRAM



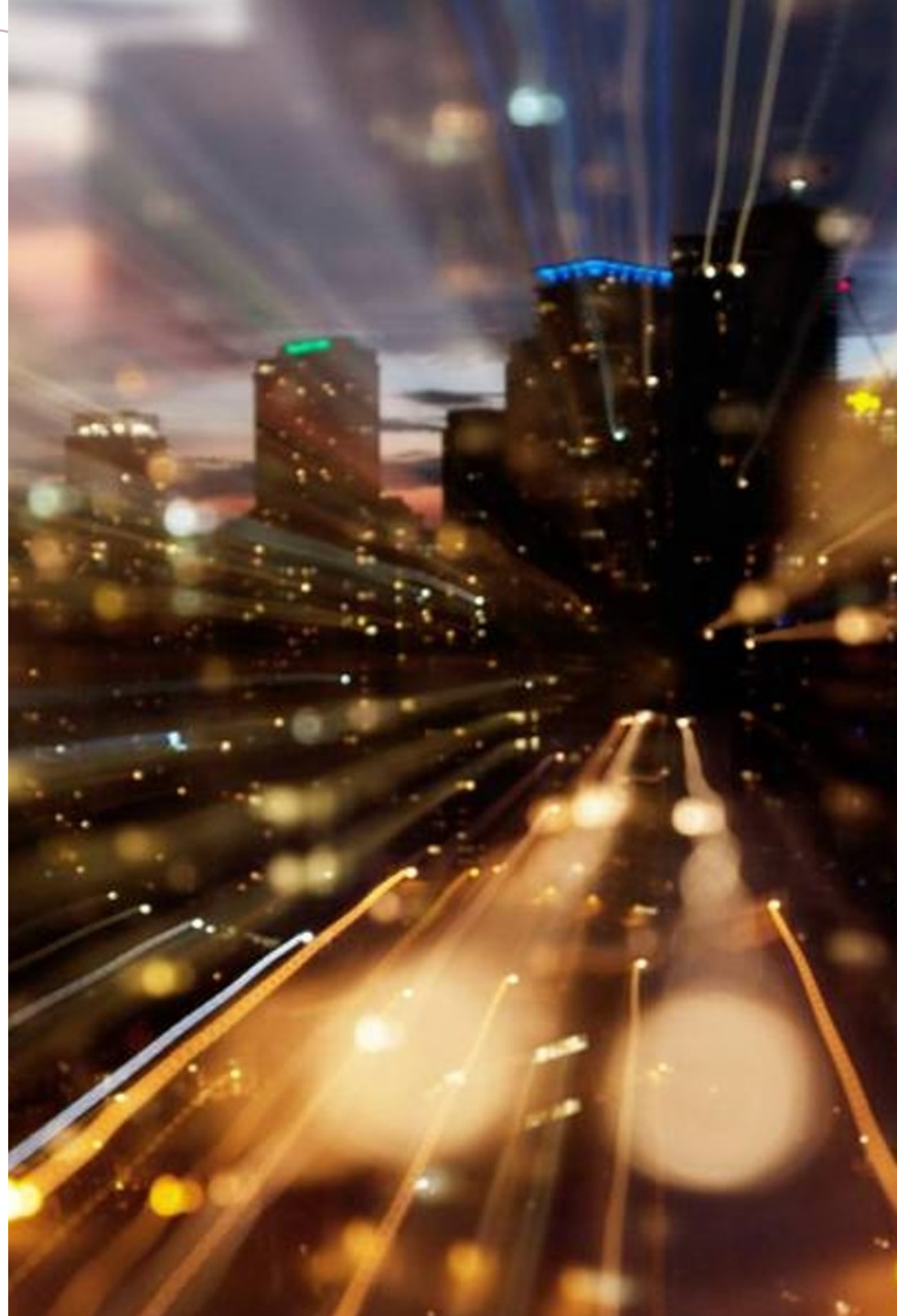
USE CASE


Student management system
Add Student
Include
Validate Student Data
View Student
Extend
Export Data
Search Student
Extend
Include
Remove Student
Extend
Confirm Removal
Generate Report
Admin


File Edit Selection View Go Run
.redhat
EXPLORER
.REDHAT
anonymousId
HotelAurora.java
Studentsmanagementsystem.java
HotelAurora.java
Studentsmanagementsystem.java
Studentsmanagementsystem.java > Studentsmanagementsystem > searchStudent()
Studentsmanagementsy
// Stor students record and scan user's input
public class Studentsmanagementsystem {
private static ArrayList<Student> students = new ArrayList<>();
private static Scanner scanner = new Scanner(System.in);
Run | Debug
public static void main(String[] args) {
int choice;
// Show Options and print the choice
do {
System.out.println("1. Add Student");
System.out.println("2. View Students");
System.out.println("3. Search Student");
System.out.println("4. Remove Student");
System.out.println("5. Exit");
System.out.print("Enter your choice: ");
choice = scanner.nextInt();
scanner.nextLine();
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PORTS
No forwarded ports. Forward a port to access your locally running services over the internet.
Forward a Port
TERMINAL
PS C:\Users\sugar\.redhat> & 'C:\Program Files\RedHat\java-21-openjdk-21.0.5.0.11-1\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sugar\AppData\Roaming\Code\User\workspaceStorage\22ea21333967414d531059d5ac6da9bb\redhat.java\jdt_ws\.redhat_cc155ca0\bin' 'Studentsmanagementsystem'
1. Add Student
2. View Students
3. Search Student
4. Remove Student
5. Exit
Enter your choice: 1
Enter student name:
Run: HotelA...
Run: Studen...
Run: Studen...
Run: Studen...
OUTLINE
TIMELINE
JAVA PROJECTS
Java: Ready
Ln 81, Col 35 Spaces: 4 UTF-8 CRLF Java

CONCLUSION

The Student and Hotel Management Systems are java-based applications intended to handle data much more efficiently because they have appropriate CRUD functionality inside without login abilities or database integrations. Proper use case diagram and class diagrams guarantee the clear line of workflows to be modularized by making these workable and user-friendly for small-level applications.





THANK YOU